

Операционные системы

Командная оболочка Midnight Commander

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

2. Процесс выполнения лабораторной работы

Работа с Midnight Commander

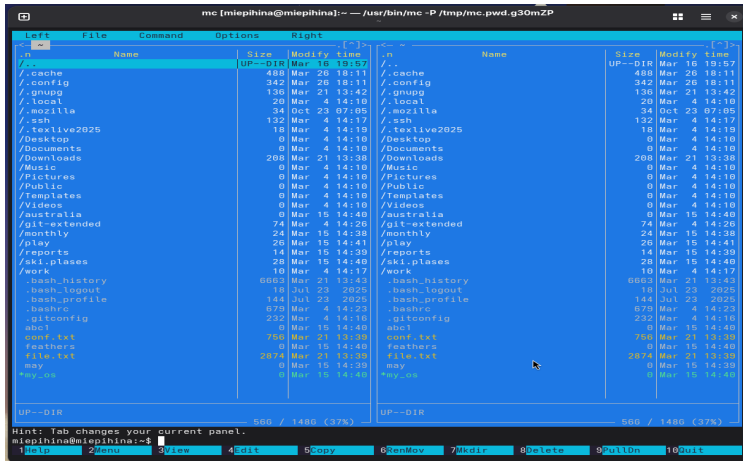


Рисунок 1: Запуск ms

Работа с Midnight Commander

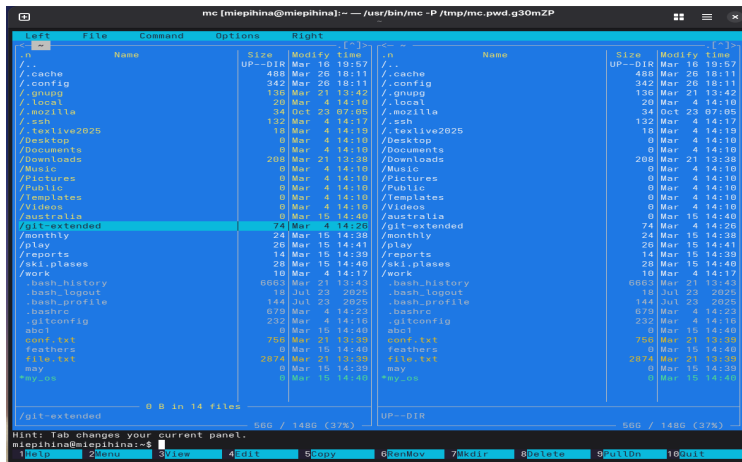


Рисунок 2: Выделение

Работа с Midnight Commander

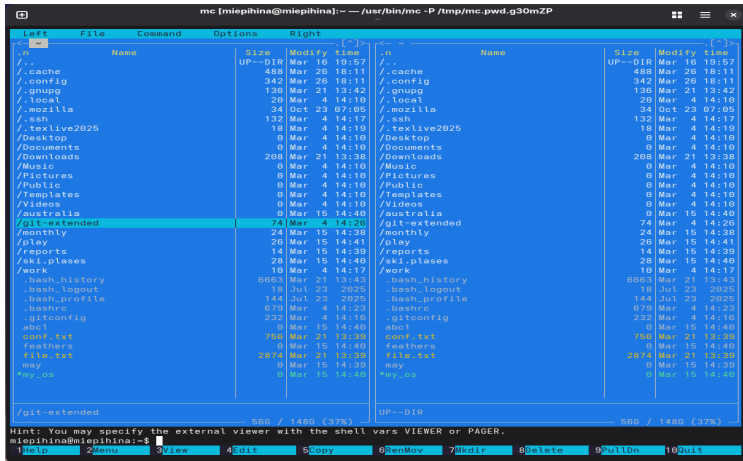


Рисунок 3: Отмена

Работа с Midnight Commander

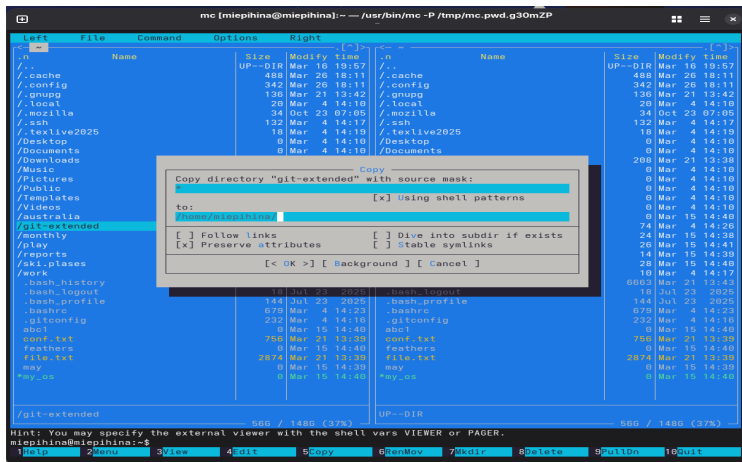


Рисунок 4: Копирование

Работа с Midnight Commander

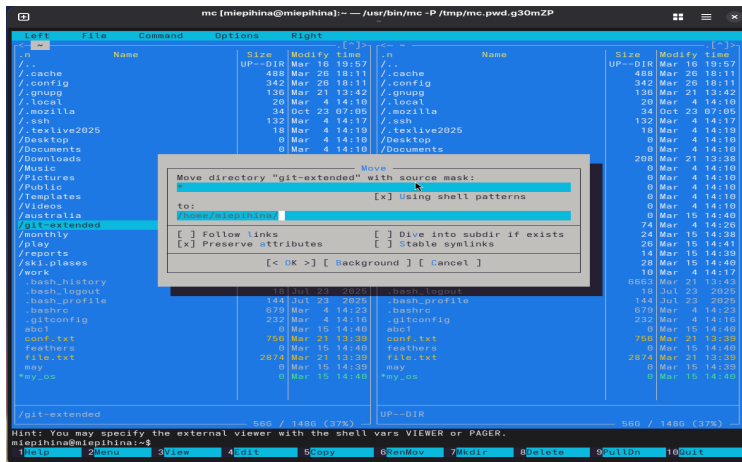


Рисунок 5: Перемещение

Работа с Midnight Commander

The screenshot shows the Midnight Commander (MC) interface. The top bar indicates the user is 'miepihina@miepihina' and the current directory is '/usr/bin/mc -P /tmp/mc.pwd.g30mZP'. The main window is divided into two panes. The left pane shows a directory listing with columns for Name, Size, and Modify time. The right pane shows detailed information for the selected file, 'git-extended'.

Left	File	Command	Options	Right
File	Name	Size	Modify time	Information
..	UP--DIR	Mar 16 19:57		Midnight Commander 4.8.33
./				File: git-extended
./cache	488	Mar 26 18:11		Location: 29h2B4EC4h
./config	342	Mar 26 18:11		Mode: drwxr-xr-x (0755)
./gnupp	136	Mar 21 13:42		Attributes: -----
./local	20	Mar 4 14:18		Links: 1
./mozilla	34	Oct 23 07:05		Owner: miepihina/miepihina
./ssh	132	Mar 4 14:17		Size: 74 (0 blocks)
./texlive2025	18	Mar 4 14:19		Changed: Mar 4 14:26
/Desktop	0	Mar 4 14:18		Modified: Mar 4 14:26
/Documents	0	Mar 4 14:18		Accessed: Mar 26 18:11
/Downloads	208	Mar 21 13:30		Filesystem: /home
/Music	0	Mar 4 14:18		Device: /dev/nvme0n1p3
/Pictures	0	Mar 4 14:18		Type: btrfs (21h)
/Public	0	Mar 4 14:18		Free space: 560 / 1486 (37%)
/Templates	0	Mar 4 14:18		No node information
/Videos	0	Mar 4 14:18		
/australia	0	Mar 15 14:48		
/git-extended	74	Mar 4 14:26		
/monthly	24	Mar 16 14:38		
/play	26	Mar 15 14:41		
/reports	14	Mar 16 14:39		
/ski.plases	28	Mar 15 14:48		
/work	10	Mar 4 14:17		
.bash_history	6663	Mar 21 13:43		
.bash_logout	18	Jul 23 2025		
.bash_profile	144	Jul 23 2025		
.bashrc	679	Mar 4 14:23		
.gitconfig	232	Mar 4 14:16		
abc1	0	Mar 15 14:48		
conf.txt	756	Mar 21 13:39		
feathers	0	Mar 15 14:48		
file.txt	2874	Mar 21 13:39		
may	0	Mar 15 14:39		
*my_os	0	Mar 15 14:48		

Hint: You may specify the external viewer with the shell vars VIEWER or PAGER.
miepihina@miepihina:~\$

1Help 2Menu 3View 4Edit 5Copy 6RenMov 7Mkdir 8Delete 9PullDn 10Quit

Рисунок 6: Информация

Работа с Midnight Commander

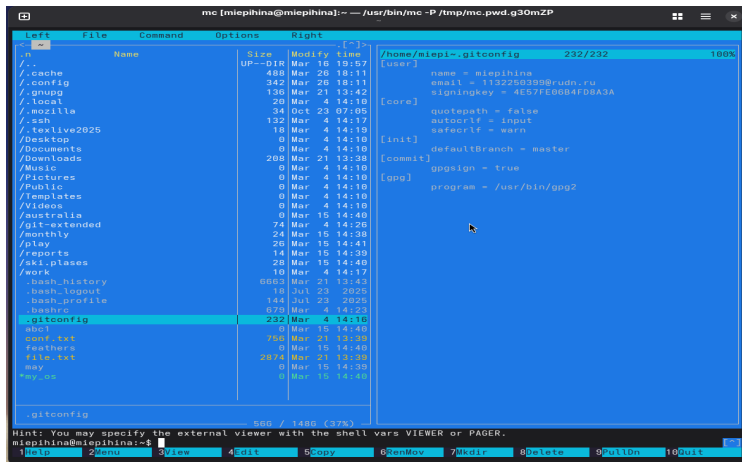


Рисунок 7: Быстрый просмотр

Работа с Midnight Commander

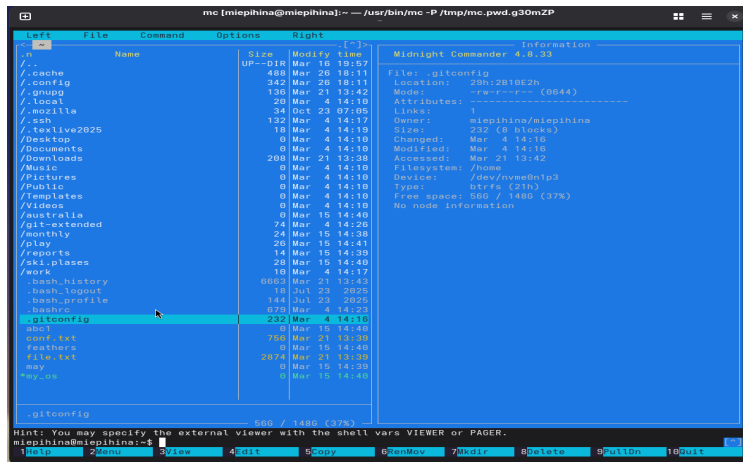


Рисунок 8: Информация

Работа с Midnight Commander

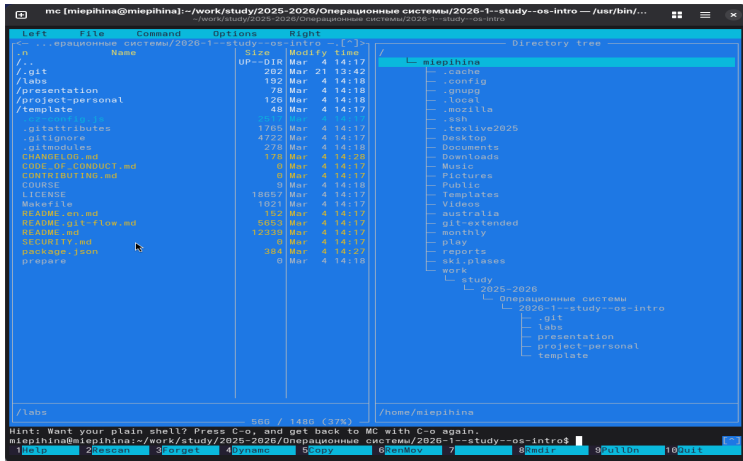
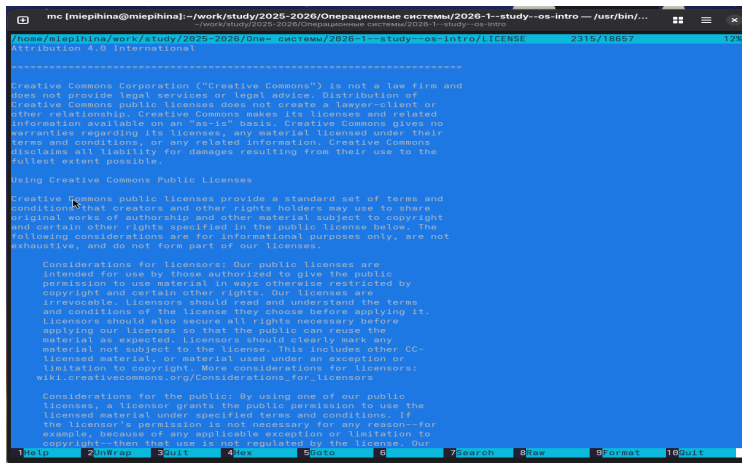


Рисунок 9: Дерево каталогов



The screenshot shows the Midnight Commander interface. The top bar displays the current directory: `mc [miepihina@miepihina]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro`. The main pane shows the contents of the file `/home/miepihina/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/LICENSE`, which is 2315/18657 bytes in size. The file content is displayed in a blue background with white text. The text includes the Creative Commons Attribution 4.0 International license notice, a disclaimer from Creative Commons Corporation, and instructions on how to use the license. The bottom status bar shows various keyboard shortcuts: 1 Help, 2 UnWrap, 3 Quit, 4 Tex, 5 Goto, 6, 7 Search, 8 Raw, 9 Format, 10 Quit.

```
mc [miepihina@miepihina]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro /usr/bin/...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro

/home/miepihina/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/LICENSE 2315/18657 12%
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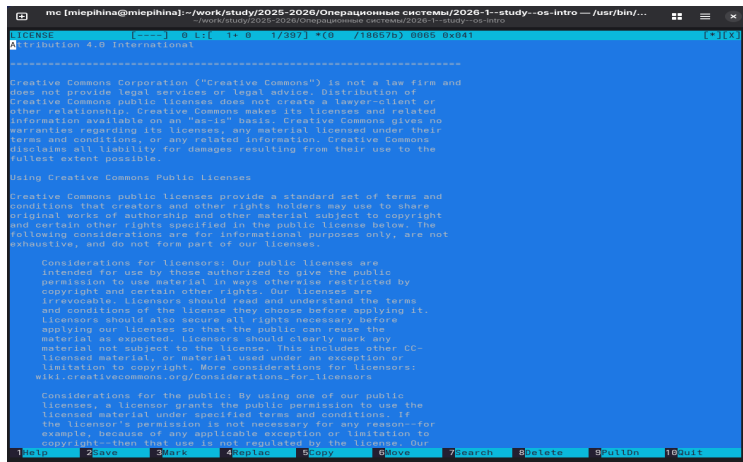
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```

Рисунок 10: Просмотр содержимого текстового файла

Работа с Midnight Commander



```
mo [miepihina@miepihina]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-Intro — /usr/bin/...
~/.work/study/2025-2026/Операционные системы/2026-1--study--os-Intro
[----] 0 L: [ 1+ 0 1/397] *(0 /18657b) 0065 0x041 [*)(X]
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1 Help 2 Save 3 Mark 4 Replace 5 Copy 6 Move 7 Search 8 Delete 9 PullDn 10 Quit
```

Рисунок 11: Отредактируем содержимое текстового файла без сохранения результатов

Работа с Midnight Commander

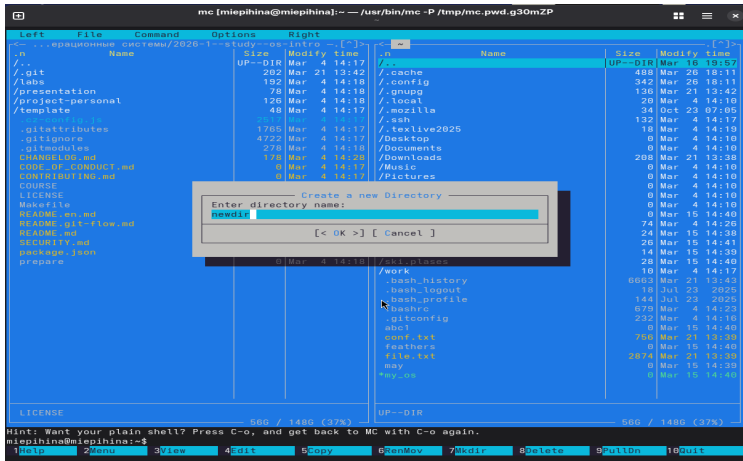


Рисунок 12: Создание каталога

Работа с Midnight Commander

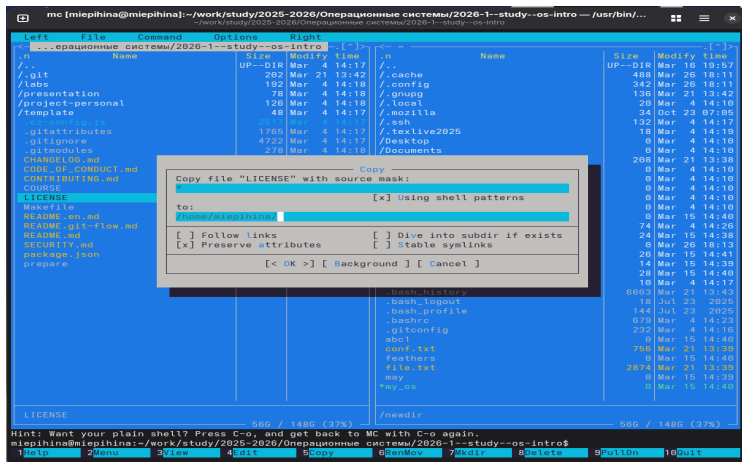


Рисунок 13: Копирование в файлов в созданный каталог

Работа с Midnight Commander

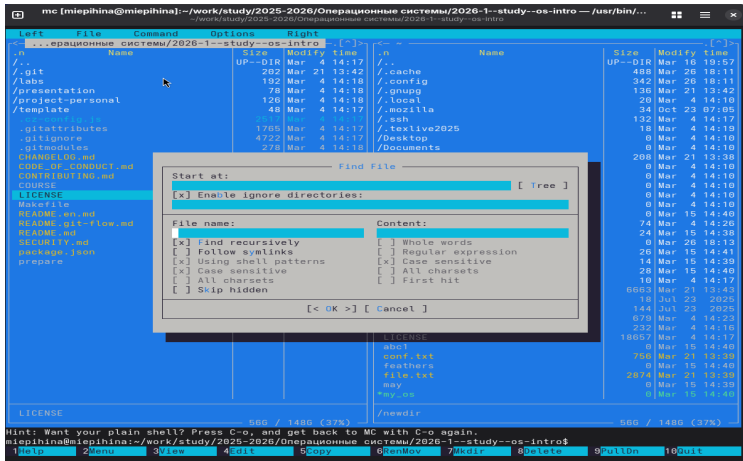


Рисунок 14: Поиск файлов

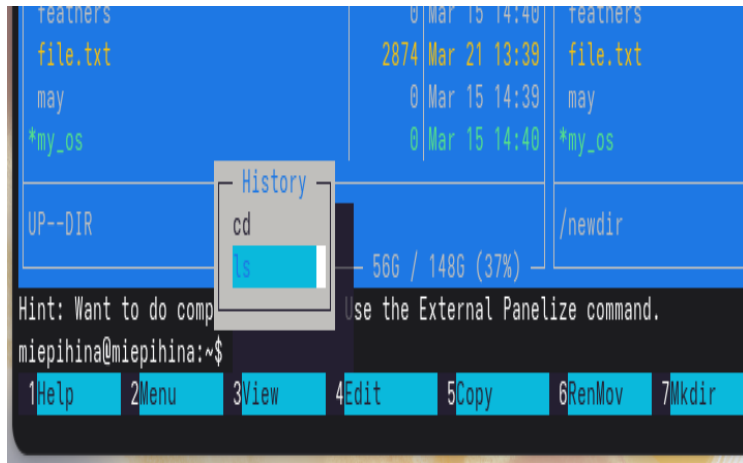


Рисунок 15: История команд

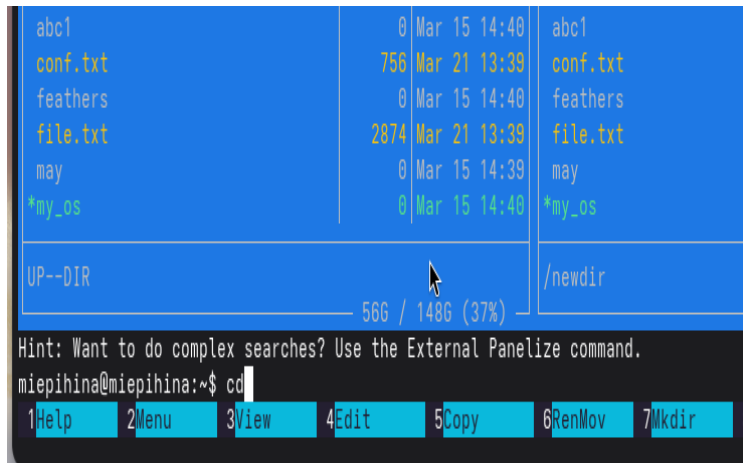
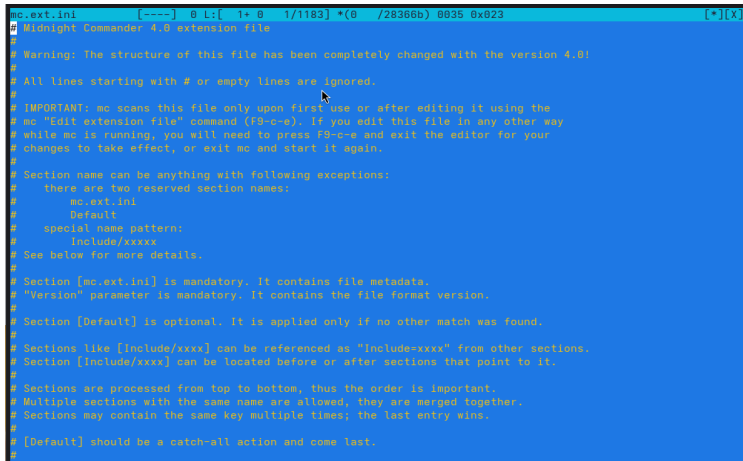


Рисунок 16: Переход в домашний каталог



```
mc.ext.ini      [---] 0 L: 1+ 0 1/1183] *(0 /28366b) 0035 0x023 [*][X]
# Midnight Commander 4.8 extension file
#
# Warning: The structure of this file has been completely changed with the version 4.8!
#
# All lines starting with # or empty lines are ignored.
#
# IMPORTANT: mc scans this file only upon first use or after editing it using the
# mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
# while mc is running, you will need to press F9-c-e and exit the editor for your
# changes to take effect, or exit mc and start it again.
#
# Section name can be anything with following exceptions:
#   there are two reserved section names:
#       mc.ext.ini
#       Default
#   special name pattern:
#       Include/xxxx
# See below for more details.
#
# Section [mc.ext.ini] is mandatory. It contains file metadata.
# "Version" parameter is mandatory. It contains the file format version.
#
# Section [Default] is optional. It is applied only if no other match was found.
#
# Sections like [Include/xxxx] can be referenced as "Include=xxxx" from other sections.
# Section [include/xxxx] can be located before or after sections that point to it.
#
# Sections are processed from top to bottom, thus the order is important.
# Multiple sections with the same name are allowed, they are merged together.
# Sections may contain the same key multiple times; the last entry wins.
#
# [Default] should be a catch-all action and come last.
```

Рисунок 17: Просмотр файла расширений

```
menu [----] 0 L:[ 1+ 0 1/380] *(0 /12204b) 0115 0x073 [X]
shell_patterns=0

#####
# %% The % character
# %f The current file (if non-local vfs, file will be copied locally and
#           %f will be full path to it)
# %p The current file
# %d The current working directory
# %s "Selected files"; the tagged files if any, otherwise the current file
# %t Tagged files
# %u Tagged files (and they are untagged on return from expand_format)
# %view Runs the commands and pipes standard output to the view command
# If %view is immediately followed by '{', recognize keywords
#   ascii, hex, proff and unform
#
# If the format letter is in uppercase, it refers to the other panel
#
# With a number followed the % character you can turn quoting on (default)
# and off. For example:
#   %f   quote expanded macro
#   %if  ditto
#   %0f  don't quote expanded macro
#####

+ ! t t
0   Do something on the current file
    CMD=%{Enter command}
    $CMD %f

+ t t
0   Do something on the tagged files
    CMD=%{Enter command}
    for i in %t ; do
```

Рисунок 18: Просмотр файла меню

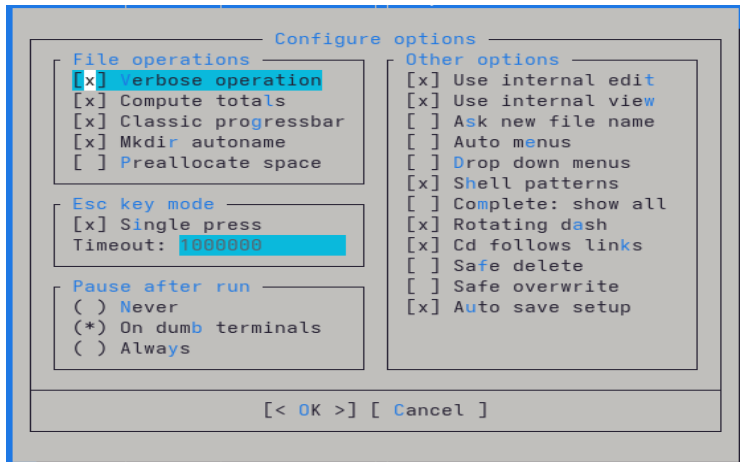


Рисунок 19: Конфигурация

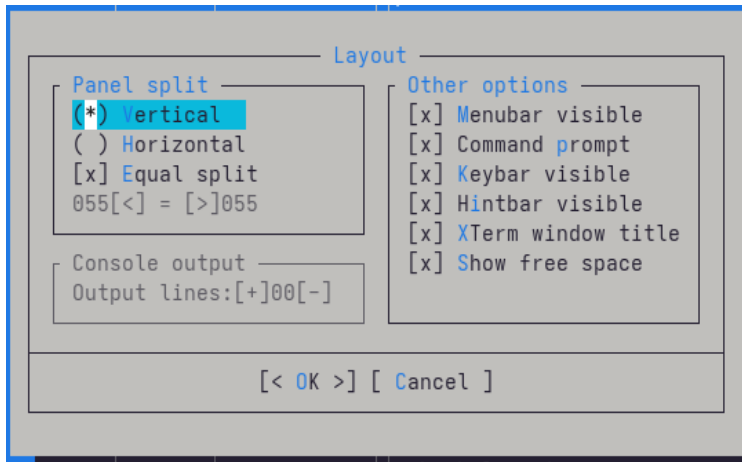


Рисунок 20: Внешний вид

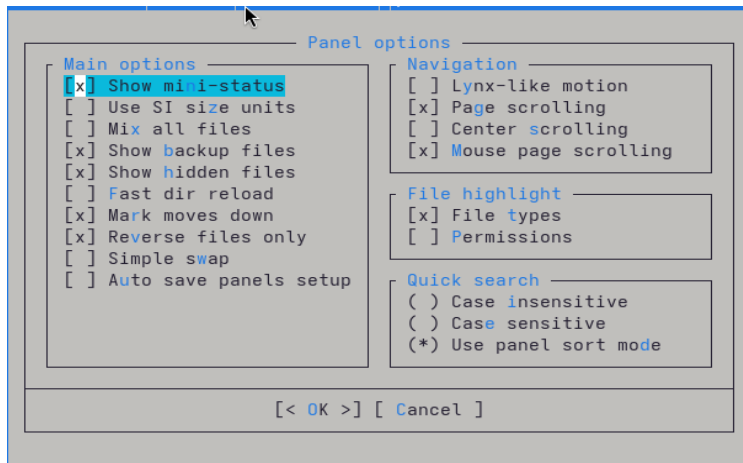


Рисунок 21: Настройки панелей

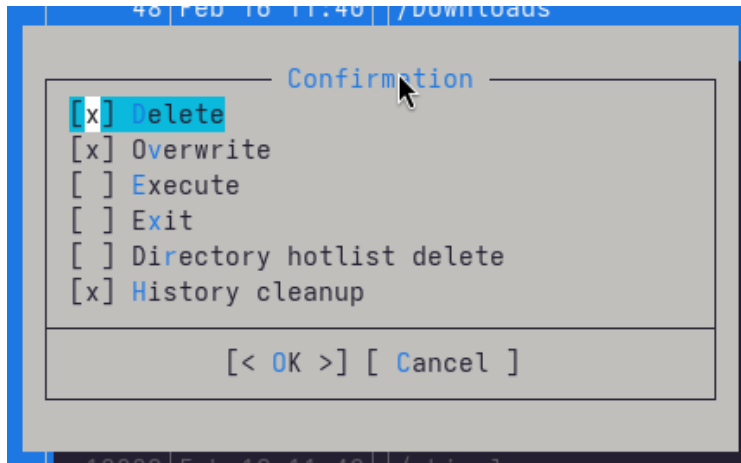


Рисунок 22: Подтверждение

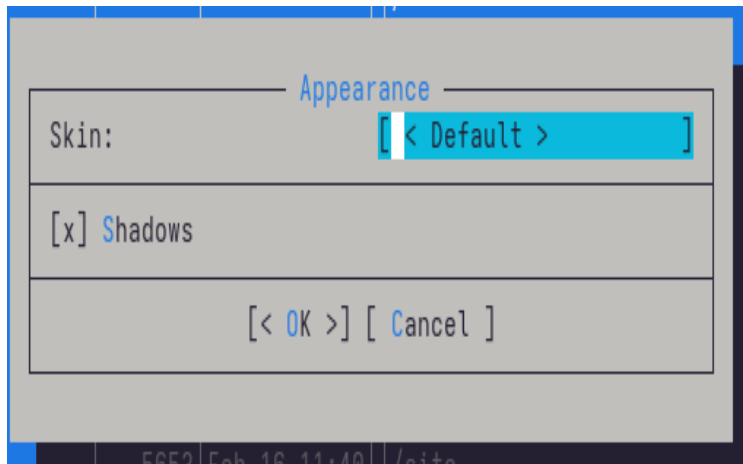


Рисунок 23: Оформление

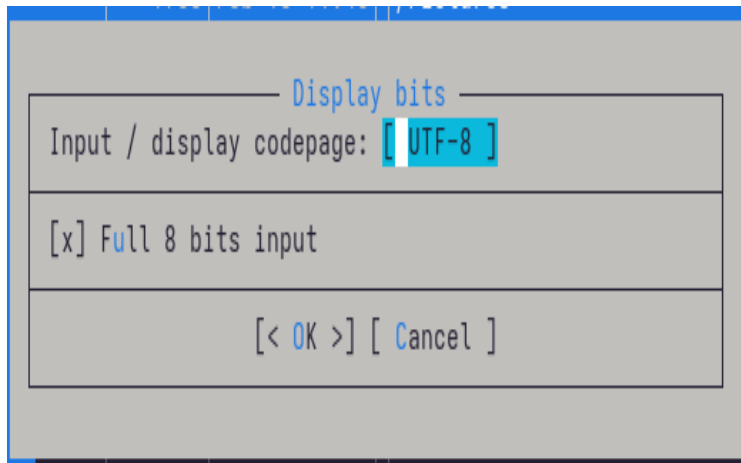


Рисунок 24: Кодировка символов

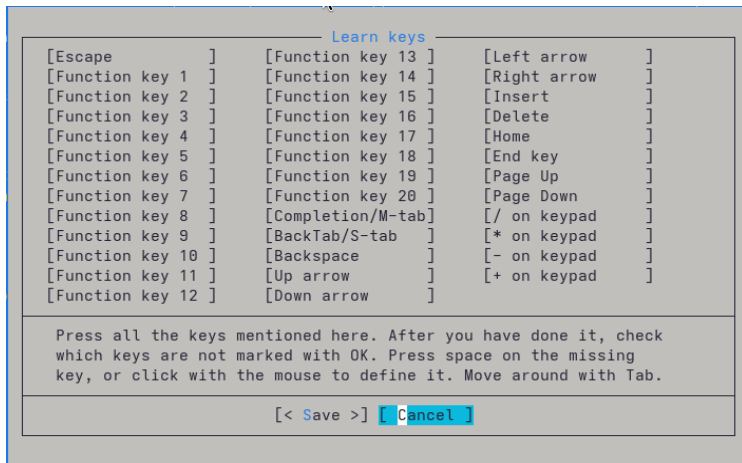
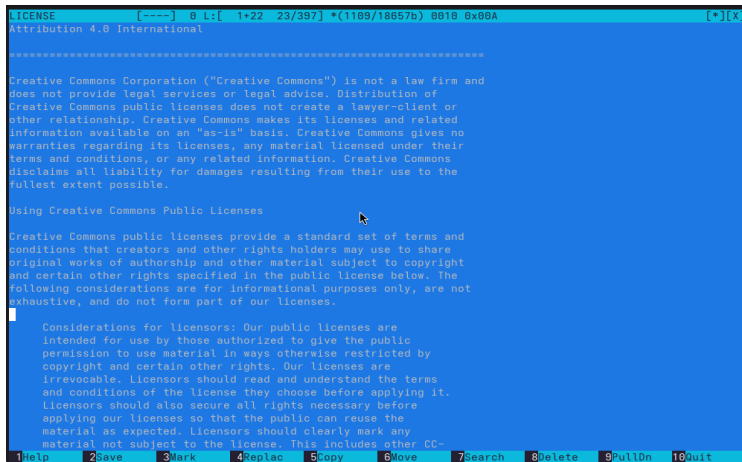


Рисунок 25: Распознавание клавиш



```
LICENSE      [-----]  8 L:[ 1+22 23/397] *(1109/18657b) 0010 0x00A      [*][X]
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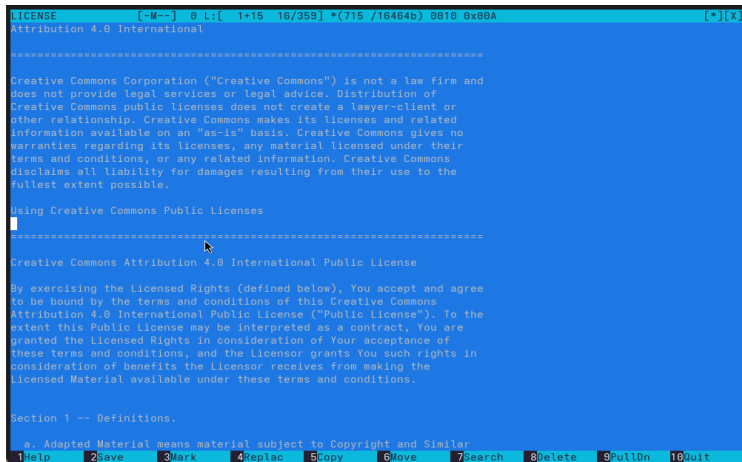
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```

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit

Рисунок 26: Файл с текстом

Работа с редактором Midnight Commander



```
LICENSE      [-M--]  8 L:[ 1+15 16/359] *(715 /16464b) 0010 0x00A      [*][x]
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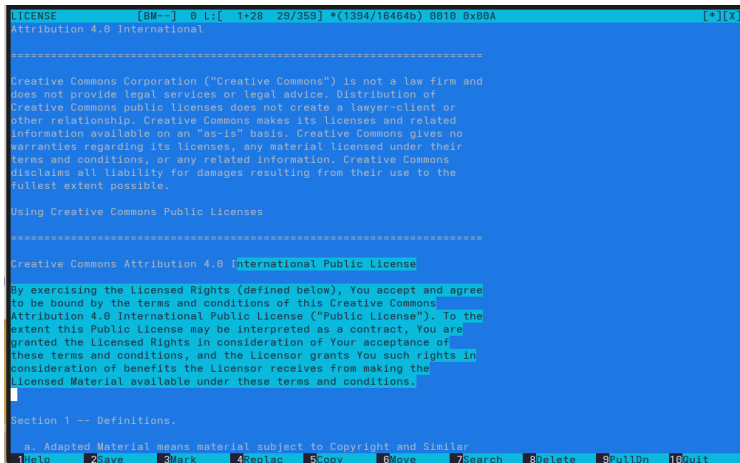
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a. Adapted Material means material subject to Copyright and Similar

1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 27: Файл с текстом

Работа с редактором Midnight Commander



```
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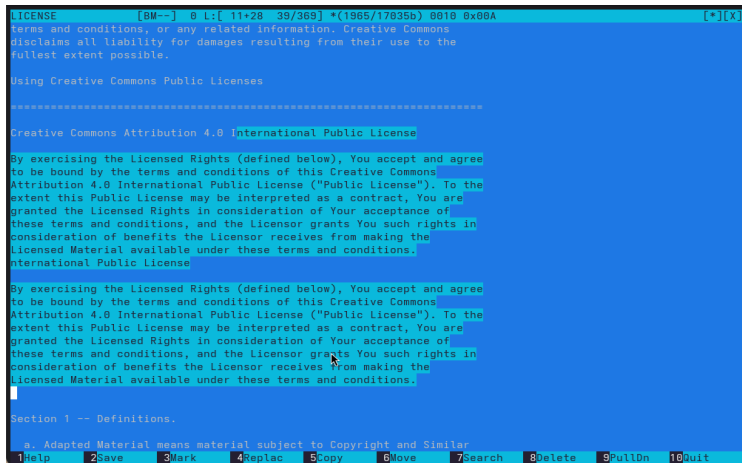
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1Help  2Save  3Mark  4Replac  5Copy  6Move  7Search  8Delete  9PullOn 10Quit
```

Рисунок 28: Копирование фрагмента



```
LICENSE      [BM--]  0 L:[ 11+28  39/369] *(1965/17035b) 0010 0x00A      [*][X]
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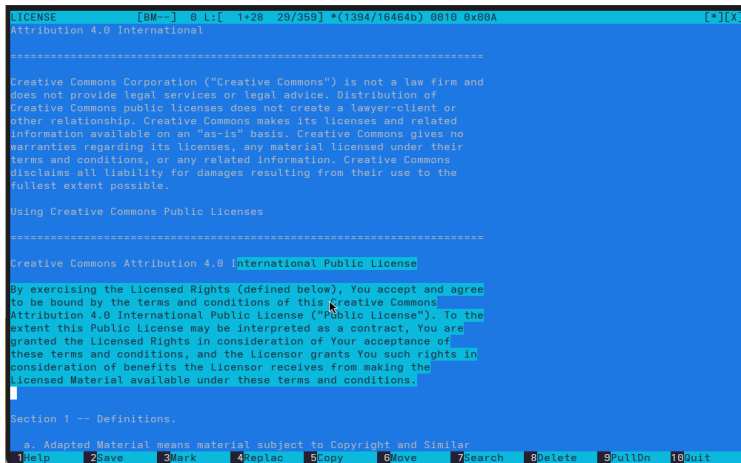
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1Help  2Save  3Mark  4Replac  5Copy  6Move  7Search  8Delete  9PullDn 10Quit
```

Рисунок 29: Сохранение

Работа с редактором Midnight Commander



```
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1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 30: Отмена

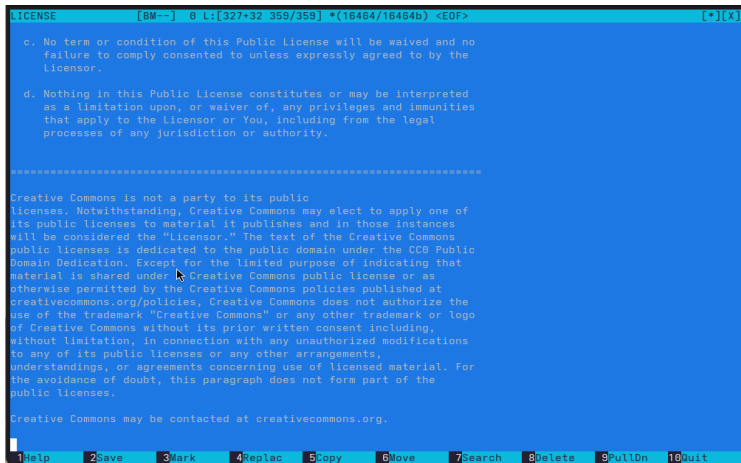
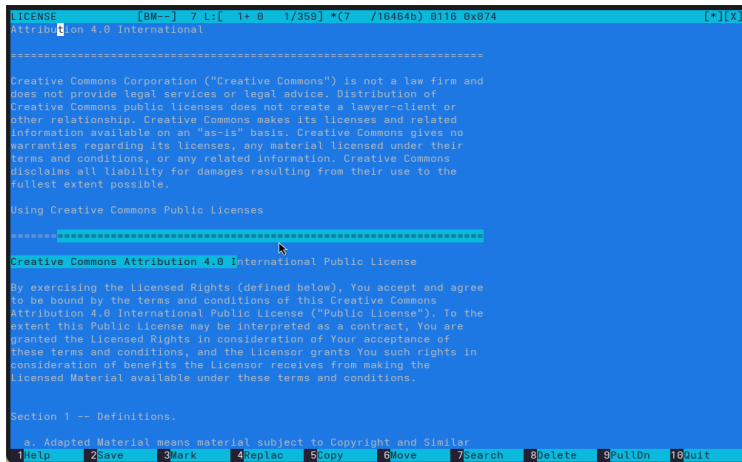


Рисунок 31: Переход в конец файла



```
LICENSE      [BM--] 7 L:[ 1+ 0 1/359] *(7 /16464b) 0116 0x074      [*][x]
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```

Рисунок 32: Переход в начало файла

```
elf.c [----] 0 L: [ 13+ 0 13/1382] *(333 /30875b) 0035 0x023 [*][X]
#include <fcntl.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <errno.h>
#include <linux/interval_tree_generic.h>
#include <objtool/builtin.h>

#include <objtool/elf.h>
#include <objtool/warn.h>

static inline u32 str_hash(const char *str)
{
    ----->return jhash(str, strlen(str), 0);
}

#define __elf_table(name)----->{elf->name##_hash}
#define __elf_bits(name)----->{elf->name##_bits}

#define __elf_table_entry(name, key) \
    ----->__elf_table(name)[hash_min(key, __elf_bits(name))]

#define elf_hash_add(name, node, key)----->----->----->----->\
    ({----->----->----->----->----->----->----->----->\
    ----->struct elf_hash_node *__node = node;----->----->----->----->\
    ----->__node->next = __elf_table_entry(name, key);----->----->----->----->\
    ----->__elf_table_entry(name, key) = __node;----->----->----->----->\
    })

static inline void __elf_hash_del(struct elf_hash_node *node,
    ----->----->----->----->-----> struct elf_hash_node **head)
{
    ----->----->----->----->----->
}
```

Рисунок 33: Файл с программой

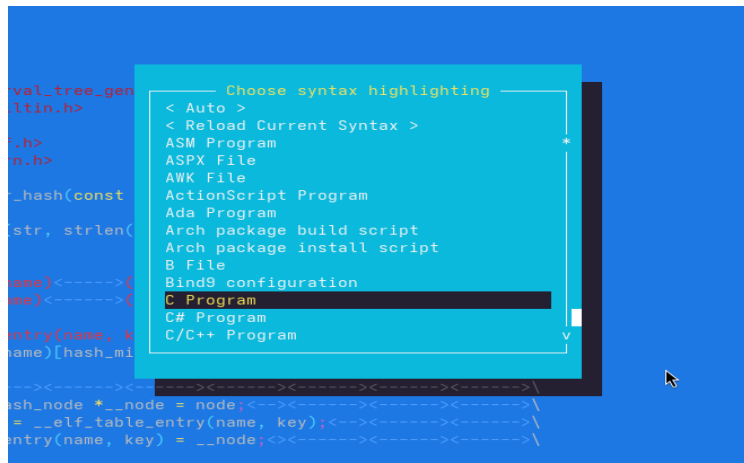


Рисунок 34: Цветовыделение синтаксиса

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.